

EDUCATION

Vellore Institute of Technology

Bhopal, India

Bachelor of Technology - Computer Science and Engineering; CGPA: 8.72

Sep 2023 - May 2027

Courses: Operating Systems, Data Structures, Analysis Of Algorithms, Artificial Intelligence, Machine Learning, Databases, Computer Networks

SKILLS

- Programming Languages:** C++ , Object-Oriented Programming (OOPS) , JavaScript (ES6+) , Problem Solving , Concurrency
- Backend Engineering:** Node.js , Express.js , MongoDB , Redis (Caching & Rate-Limiting) , RESTful APIs , Auth & Authorization
- Core Concepts & Data Structures:** Data Structures, Algorithms, OOP, Database Management Systems (DBMS), Operating Systems, Computer Networks
- Databases & Caching:** MySQL / SQL, MongoDB, Redis, Mongoose, Indexing, Query Optimization, Joins
- Software Design & Modeling:** System Design Basics, UML Modeling, ER Diagrams, Data Flow Diagrams (DFD)
- DevOps & Developer Tools:** Git, GitHub, AWS Deployment, Postman, MS-SQL, Firebase, Excel, Render, Vercel, Docker

PROJECTS

- DPI — Deep Packet Inspection System GitHub**
 - Developed a high-performance Deep Packet Inspection engine in C++ using `libpcap` to ingest, decode, and analyze network traffic from PCAP trace files.
 - Built a byte-level protocol parsing pipeline across Data Link (Ethernet), Network (IPv4/IPv6), and Transport (TCP/UDP) layers to perform deep payload inspection.
 - Implemented stateful flow tracking using a 5-Tuple model (`src_ip`, `dst_ip`, `src_port`, `dst_port`, `protocol`) to track TCP state transitions (SYN, ESTABLISHED, FIN/RST).
 - Integrated a TLS Client Hello parser to extract SNI fields from encrypted HTTPS traffic, classifying domain applications (YouTube, Facebook) without decryption.
 - Architected a multi-threaded execution pipeline featuring a load balancer and worker pools using thread-safe queues, mutexes, and condition variables.
 - Engineered a rule-based firewall filtering engine supporting pattern-matching on IPs, ports, and domains alongside real-time traffic telemetry reporting.**Tech:** C++, Multithreading, TCP/IP, TLS Handshake Parsing, PCAP, Concurrency Control, Data Structures, System Design
- FinLedger Pro — Backend Banking Infrastructure GitHub | Live Demo**
 - Architected a financial backend microservice using Node.js/Express, implementing a strict double-entry ledger system with atomic `CREDIT` and `DEBIT` entries.
 - Enforced schema-level immutability using Mongoose write/update/delete blocking hooks, dynamically computing balances via MongoDB aggregation pipelines.
 - Integrated Redis via the Cache-Aside pattern to cache dynamic balance calculations, bypassing DB aggregations and reducing latency from ~45ms to < 1ms.
 - Implemented custom Redis-backed distributed rate-limiting middleware to defend authentication (`/api/auth`) and transaction (`/api/transactions`) endpoints.
 - Guaranteed ACID transaction compliance across multi-document writes using MongoDB sessions (`startSession`) with automatic abort/rollback capabilities.
 - Prevented double-spending vulnerabilities across network retries by enforcing unique UUID-based `idempotencyKey` verification on payment workflows.
 - Implemented JWT cookie authentication paired with MongoDB TTL-indexed token blacklisting (`expireAfterSeconds: 3 days`) for instant session revocation.
 - Built asynchronous email notification services with Nodemailer and OAuth2.0 authentication to dispatch transaction receipts without blocking response cycles.**Tech:** Node.js, Express.js, MongoDB, Mongoose, Redis, JWT, Nodemailer, RESTful API Design, System Architecture

CERTIFICATIONS

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|---|-----------|
| Google IT Support Professional Certificate – Google (Google Skills) | Jan 2026 |
| The Bits and Bytes of Computer Networking – Google (Coursera) | Dec 2025 |
| AWS Technical Essentials – AWS Educate (Amazon) | June 2025 |
| Operating Systems and Systems Programming – Cursa | Apr 2025 |
| Frontend Development Libraries – freeCodeCamp | Sep 2024 |
| Introduction to Machine Learning – NPTEL (IIT) | Sep 2025 |
| Market Analysis – NPTEL (IIT) | May 2026 |

POSITIONS OF RESPONSIBILITY

- Cultural Representative – Pahadi Club, VIT Bhopal, India
- April 2025 – Present
- Led cultural events, fostering teamwork, collaboration, and cross-functional coordination across campus initiatives.